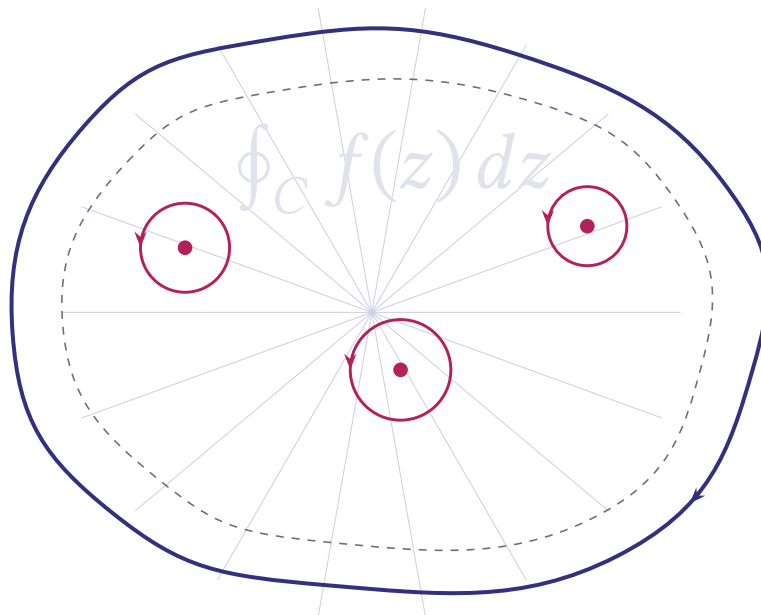


Integration of Complex Functions

انتگرال توابع مختلط



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5.1 The Definite Integral and the Path of Integration

The integral of complex functions, like that of real functions, appears in two forms:

- the indefinite integral, which amounts to finding an antiderivative;
- the definite integral, known as the integral along a path.

In the setting of complex functions the definite integral is also called the path integral or *line integral*, and is written

$$\int_C f(z) dz$$

C is called the path of integration.

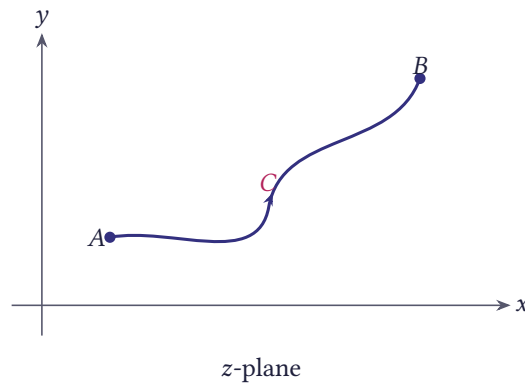


Figure 5.1 A path of integration C from the point A to the point B in the z -plane

■ The Path of Integration

A path of integration in the complex plane consists of a starting point, an intermediate route and an end point. In many cases the best way to describe a path is to use an auxiliary variable which, as it increases, traverses the path from start to end in the desired direction.

$$z(t) = x(t) + iy(t), \quad a \leq t \leq b, \quad t, a, b \in \mathbb{R}$$

$$z(a) = A, \quad z(b) = B, \quad z, A, B \in \mathbb{C}$$

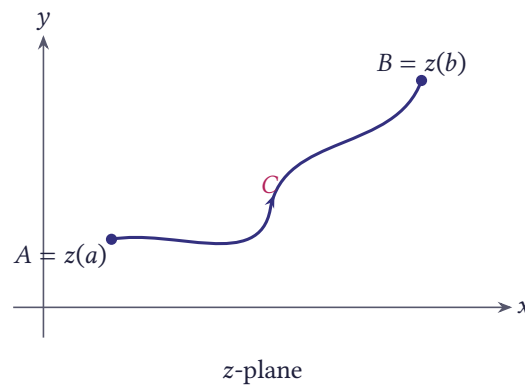


Figure 5.2 The path C , directed from $A = z(a)$ to $B = z(b)$

Example 5.1

$$z(t) = t + i2t, \quad 1 \leq t \leq 2$$

This path is the line segment from the point $1 + 2i$ to the point $2 + 4i$.

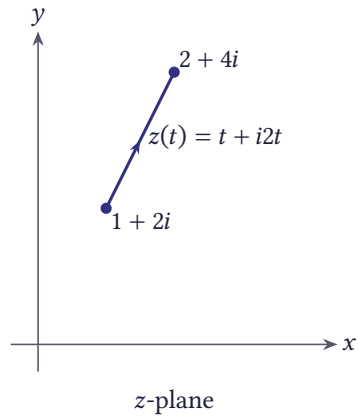


Figure 5.3 The path $z(t) = t + i2t$, $1 \leq t \leq 2$

Example 5.2

$$z(t) = \cos t + i \sin t, \quad 0 \leq t \leq 2\pi$$

$$z(t) = e^{it}, \quad 0 \leq t \leq 2\pi$$

This path is the unit circle, traversed counterclockwise.

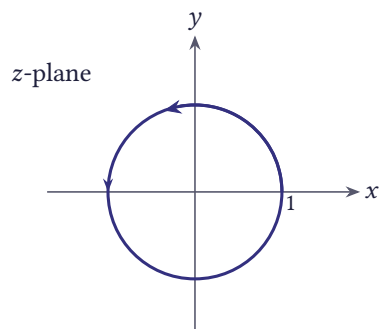


Figure 5.4 The path $z(t) = e^{it}$, $0 \leq t \leq 2\pi$ (the unit circle)

■ The Derivative of the Path

$$\dot{z}(t) = \frac{d}{dt}z(t) = \lim_{\Delta t \rightarrow 0} \frac{z(t + \Delta t) - z(t)}{\Delta t} = \frac{d}{dt}x(t) + i \frac{d}{dt}y(t) = \dot{x}(t) + i\dot{y}(t)$$

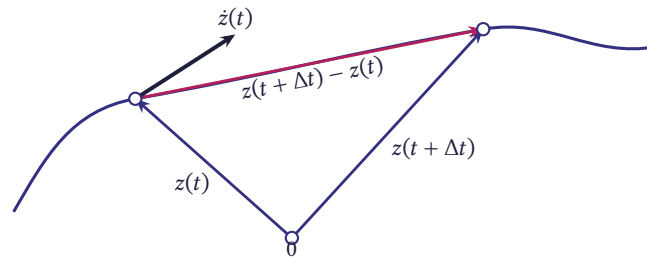


Figure 5.5 The derivative of the path $\dot{z}(t)$ as the limit of $\frac{z(t+\Delta t) - z(t)}{\Delta t}$

5.2 The Definite Integral of Complex Functions

$$\begin{aligned} \int_C f(z) dz &= \int_C (u + iv)(dx + i dy) = \int_C (u dx - v dy) + i \int_C (v dx + u dy) \\ \int_C f(z) dz &= \int_t f(z(t)) \dot{z}(t) dt \end{aligned}$$

Definition 5.3 ■ Steps for computing a complex definite integral along a path

1. Describe the path parametrically: $z(t) = x(t) + iy(t)$, $a \leq t \leq b$.
2. Compute the derivative of $z(t)$ with respect to t : $\dot{z}(t) = \dot{x}(t) + i\dot{y}(t)$.
3. Substitute $z(t)$ into $f(z)$.
4. Integrate $f(z(t))\dot{z}(t)$ with respect to t for $a \leq t \leq b$.

Example 5.4

Compute the integral of the function $f(z) = \bar{z}$ along the path shown.

$$z(t) = t + it^2, \quad -1 \leq t \leq 1, \quad \dot{z}(t) = 1 + i2t$$

$$f(z) = \bar{z} = x - iy, \quad f(z(t)) = \bar{z}(t) = t - it^2$$

$$\begin{aligned} \int_{-1+i}^{1+i} \bar{z} dz &= \int_{t=-1}^1 (t - it^2)(1 + i2t) dt \\ &= \int_{t=-1}^1 [(t + 2t^3) + i(2t^2 - t^2)] dt = \int_{t=-1}^1 [(t + 2t^3) + it^2] dt \\ &= \left(\frac{1}{2}t^2 + \frac{2}{4}t^4 \right) + i \times \frac{1}{3}t^3 \Big|_{-1}^1 = 0 + i \frac{2}{3} \end{aligned}$$

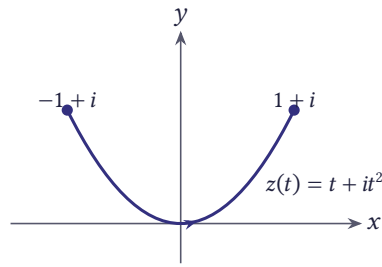


Figure 5.6 The path $z(t) = t + it^2$, $-1 \leq t \leq 1$, from $-1 + i$ to $1 + i$

Example 5.5

Compute the integral of the function $f(z) = \bar{z}$ along the path shown.

$$z(t) = t + i, \quad -1 \leq t \leq 1, \quad \dot{z}(t) = 1$$

$$f(z) = \bar{z} = x - iy, \quad f(z(t)) = \bar{z}(t) = t - i$$

$$\int_{-1+i}^{1+i} \bar{z} dz = \int_{t=-1}^1 (t - i)(1) dt = \left(\frac{t^2}{2} - it \right) \Big|_{-1}^1 = 0 - i2$$

So for this complex function the value of the integral depends on the chosen path.

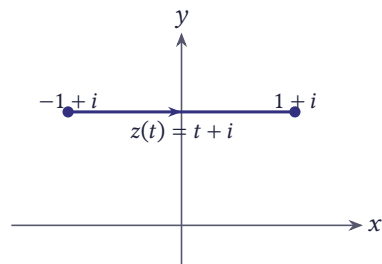


Figure 5.7 The path $z(t) = t + i$, $-1 \leq t \leq 1$, from $-1 + i$ to $1 + i$

■ Some Properties of the Definite Integral

$$\int_C [k_1 f_1(z) + k_2 f_2(z)] dz = k_1 \int_C f_1(z) dz + k_2 \int_C f_2(z) dz \quad (\text{linearity})$$

$$\int_{z_0}^Z f(z) dz = - \int_Z^{z_0} f(z) dz \quad (\text{reversing the direction of integration})$$

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz \quad (\text{splitting the path of integration})$$

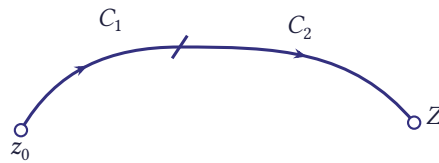


Figure 5.8 A path C from z_0 to Z split into two pieces C_1 and C_2

5.3 Simple Closed Curves and Simply Connected Domains

Definition 5.6 ■ Simple closed curve

A simple closed curve is a closed curve that does not intersect itself.

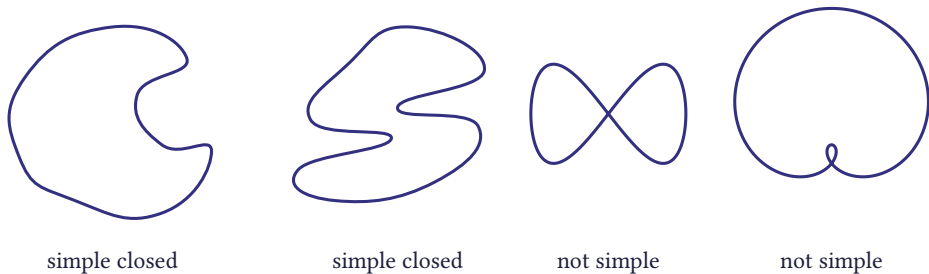


Figure 5.9 Simple closed curves and closed curves that are not simple

Definition 5.7 ■ Simply connected domain

A domain D is simply connected if, whenever a simple closed curve lies entirely in it, all the points inside the curve belong to this domain.

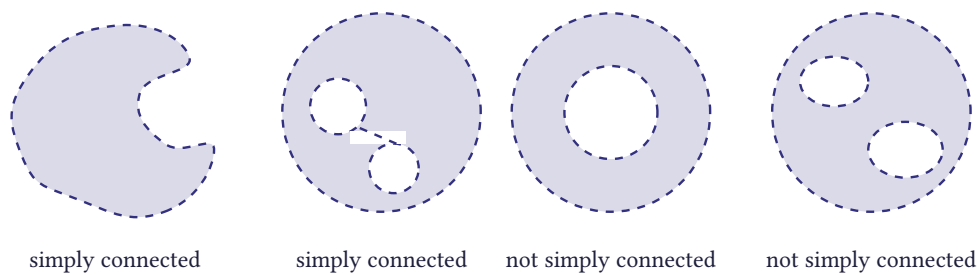


Figure 5.10 Simply connected and not simply connected domains

5.4 Integration by Antiderivative

Theorem 5.8

Let the complex function $f(z)$ be analytic in the simply connected domain D . If $F(z)$ is an antiderivative of $f(z)$, i.e. $F'(z) = f(z)$, then

$$\int_{z_0}^{z_1} f(z) dz = F(z_1) - F(z_0)$$

Example 5.9

$$\int_0^{1+i} z^2 dz = \frac{1}{3} z^3 \Big|_0^{1+i} = \frac{1}{3} (1+i)^3 = \frac{2}{3} (-1+i)$$

Example 5.10

$$\int_{-\pi i}^{\pi i} \cos z dz = \sin z \Big|_{-\pi i}^{\pi i} = 2 \sin(\pi i) = 2i \sinh \pi = 23.097 i$$

Example 5.11

$$\int_{8+\pi i}^{8-3\pi i} e^{z/2} dz = 2e^{z/2} \Big|_{8+\pi i}^{8-3\pi i} = 2(e^{4-3\pi i/2} - e^{4+\pi i/2}) = 0$$

Example 5.12

$$\int_{-i}^i \frac{dz}{z} = \ln i - \ln(-i) = \frac{i\pi}{2} - \left(-\frac{i\pi}{2}\right) = i\pi$$

5.5 Integrals of Complex Functions Along Closed Paths

The integral along a simple closed path C , written

$$\oint_C f(z) dz$$

is of great importance.

Theorem 5.13 ■ Cauchy's integral theorem

If the function $f(z)$ is analytic in the simply connected domain D , then for every simple closed path C in this domain,

$$\oint_C f(z) dz = 0$$

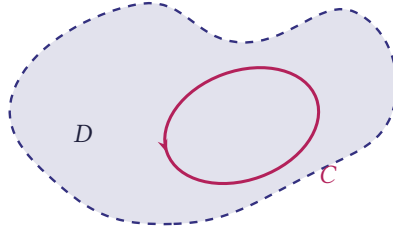


Figure 5.11 A simple closed path C in the simply connected domain D

Proof.

As we saw earlier, since the complex function is analytic, we have

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\oint_C f(z) dz = \oint_C (u + iv)(dx + i dy) = \oint_C (u dx - v dy) + i \oint_C (v dx + u dy)$$

By Green's theorem (which you learned in Mathematics II),

$$\oint_C (L dx + M dy) = \iint_R \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) dx dy$$

Hence

$$\oint_C (u dx - v dy) + i \oint_C (v dx + u dy) = \iint_R \underbrace{\left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)}_0 dx dy + i \iint_R \underbrace{\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right)}_0 dx dy = 0$$

where the first term vanishes because of the second Cauchy–Riemann condition and the second because of the first. ■

Example 5.14

If the function is analytic in the whole complex plane, the integral along any closed path is zero:

$$\oint_C \cos z dz = 0, \quad \oint_C e^z dz = 0, \quad \oint_C z^n dz = 0, \quad n = 1, 2, 3, \dots$$

Example 5.15

If the function has non-analytic points that lie neither on the path nor inside C , the integral is still zero:

$$\oint_{|z|=1} \sec z \, dz, \quad \oint_{|z|=1} \frac{dz}{z^2 + 4}$$

(the path is the unit circle.)

Example 5.16

If the function is not analytic, the integral along a closed path **may** be nonzero:

$$\oint_{|z|=1} \bar{z} \, dz, \quad z(t) = \cos t + i \sin t = e^{it}, \quad \dot{z}(t) = ie^{it}$$

$$\oint_{|z|=1} \bar{z} \, dz = \int_0^{2\pi} e^{-it} ie^{it} dt = 2\pi i$$

Example 5.17

If the function is not analytic on the path or inside it, the integral along a closed path **may** be zero:

$$\oint_C \frac{1}{z^2} dz = 0$$

Exercise 5.18

Prove that $\oint_C \frac{1}{z^2} dz = 0$.

Consequence 1

The connectedness of the domain is essential in Cauchy's integral theorem.

Consequence 2

If $f(z)$ is analytic in the simply connected domain D , the definite integral does not depend on the choice of path.

$$\int_{C_1} f(z) dz + \int_{C_2^*} f(z) dz = 0 \implies \int_{C_1} f(z) dz = - \int_{C_2^*} f(z) dz \implies \int_{C_1} f(z) dz = \int_{C_2} f(z) dz$$

Hence the integral from the starting point to the end point does not depend on the chosen path.

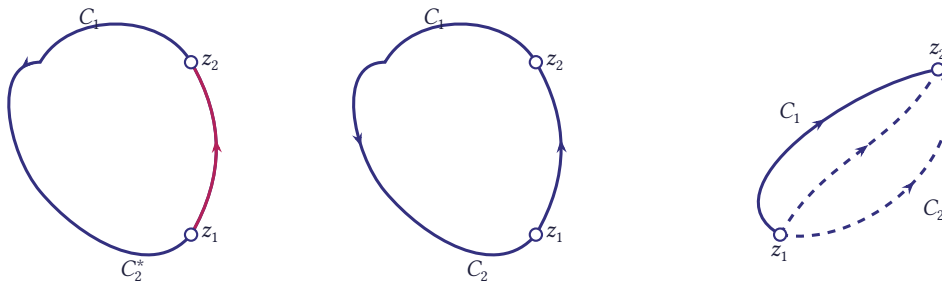


Figure 5.12 The paths C_1 , C_2 and C_2^* (C_2 with reversed direction) between z_1 and z_2

■ A Very Important Example

$$\oint_C \frac{dz}{z - z_0} = ? \quad C : z(t) = z_0 + re^{it}$$

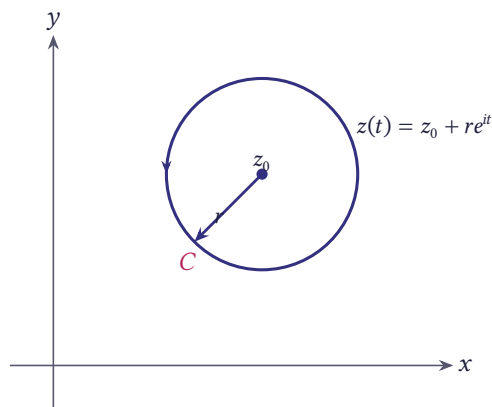


Figure 5.13 A circle C with centre z_0 and radius r

$$z(t) = z_0 + re^{it}, \quad \dot{z}(t) = rie^{it}$$

$$\oint_C \frac{dz}{z - z_0} = \int_0^{2\pi} \frac{rie^{it}}{re^{it}} dt = \int_0^{2\pi} i dt = 2\pi i$$

Key result 5.19

$$\oint_C \frac{dz}{z - z_0} = 2\pi i$$

5.6 Cauchy's Integral Formula

Theorem 5.20 ■ Cauchy's integral formula

If the function $f(z)$ is analytic in the simply connected domain D and z_0 is a point inside C , then

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz = f(z_0)$$

Proof.

As we saw earlier, since the function $\frac{f(z)}{z - z_0}$ is analytic inside the connected domain Γ , we have

$$\begin{aligned} \oint_{\Gamma} \frac{f(z)}{z - z_0} dz &= 0 = \int_{ACA'} \frac{f(z)}{z - z_0} dz + \int_{A'B'} \frac{f(z)}{z - z_0} dz + \int_{B'C'B} \frac{f(z)}{z - z_0} dz + \int_{BA} \frac{f(z)}{z - z_0} dz \\ \lim_{\substack{A \rightarrow A' \\ B \rightarrow B'}} \int_{BA} \frac{f(z)}{z - z_0} dz &= - \int_{A'B'} \frac{f(z)}{z - z_0} dz \end{aligned}$$

Consequently (the two integrals along the connecting segments cancel)

$$\lim_{\substack{A \rightarrow A' \\ B \rightarrow B'}} \int_{ACA'} \frac{f(z)}{z - z_0} dz + \int_{B'C'B} \frac{f(z)}{z - z_0} dz = 0 \implies \oint_C \frac{f(z)}{z - z_0} dz = \oint_{C'} \frac{f(z)}{z - z_0} dz$$

(here C' is traversed counterclockwise). Continuing:

$$\begin{aligned} \oint_C \frac{f(z)}{z - z_0} dz &= \oint_{C'} \frac{f(z)}{z - z_0} dz = \oint_{C'} \frac{f(z) - f(z_0) + f(z_0)}{z - z_0} dz \\ &= \oint_{C'} \frac{f(z) - f(z_0)}{z - z_0} dz + \oint_{C'} \frac{f(z_0)}{z - z_0} dz \\ &= \oint_{C'} g(z) dz + f(z_0) \oint_{C'} \frac{1}{z - z_0} dz = 0 + 2\pi i f(z_0) = 2\pi i f(z_0) \end{aligned}$$

where

$$g(z) \triangleq \begin{cases} \frac{f(z) - f(z_0)}{z - z_0} & z \neq z_0 \\ f'(z_0) & z = z_0 \end{cases}$$

is analytic inside C' . Therefore

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz = f(z_0)$$

■

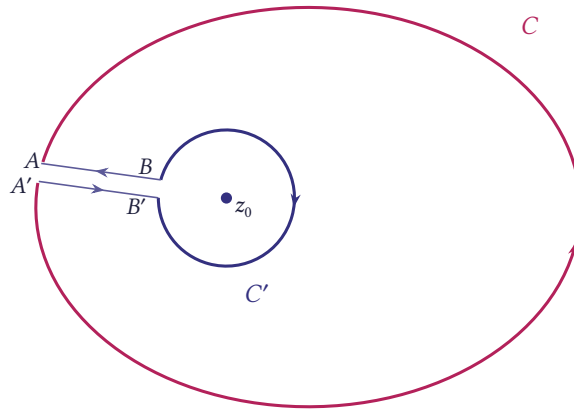


Figure 5.14 The path Γ in the proof of Cauchy's integral formula: the curve C , the small circle C' centred at z_0 , and the two connecting segments AA' and BB'

5.7 Application: Evaluating Real Integrals

Example 5.21

Evaluate the integral $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ using Cauchy's integral formula.

Consider the function $g(z) = \frac{1}{1+z^2}$. Using the path shown in the figure we find the integral along the closed path.

$$\oint_C \frac{dz}{1+z^2} = \oint_C \left[\frac{1}{z-i} \right] dz = 2\pi i \frac{1}{i+i} = \pi, \quad f(z) = \frac{1}{z+i}, \quad z_0 = i$$

On the other hand

$$\oint_C \frac{dz}{1+z^2} = \int_{-R}^R \frac{dx}{1+x^2} + \int_0^\pi \frac{iRe^{it} dt}{1+R^2e^{i2t}}$$

In the expression above, the first term as $R \rightarrow \infty$ is the required value.

$$\oint_C \frac{dz}{1+z^2} = \pi = \int_{-\infty}^{\infty} \frac{dx}{1+x^2} + I, \quad I = \lim_{R \rightarrow \infty} \int_0^\pi \frac{iRe^{it} dt}{1+R^2e^{i2t}} = 0$$

Therefore

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$$

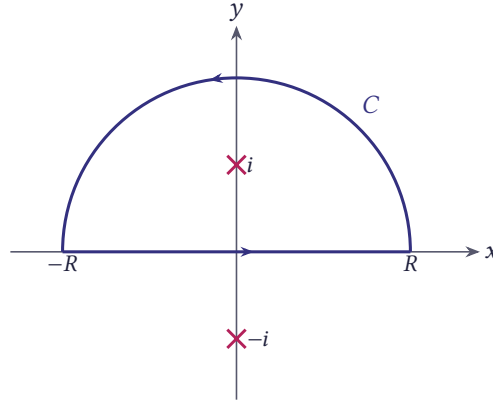


Figure 5.15 The semicircular path in the upper half-plane; the poles are at $z = \pm i$

Example 5.22

Evaluate the integral $\int_{-\infty}^{\infty} \frac{\cos \omega x}{1+x^2} dx$ using Cauchy's integral formula ($\omega > 0$).

Consider the function $g(z) = \frac{e^{i\omega z}}{1+z^2}$. Using the path shown in the figure (the same semicircle as before) we find the integral along the closed path.

$$\oint_C \frac{e^{i\omega z}}{1+z^2} dz = \oint_C \frac{\left[\frac{e^{i\omega z}}{z-i} \right]}{z+i} dz = 2\pi i \frac{e^{i\omega i}}{i+i} = \pi e^{-\omega}, \quad f(z) = \frac{e^{i\omega z}}{z+i}, \quad z_0 = i$$

On the other hand

$$\oint_C \frac{e^{i\omega z}}{1+z^2} dz = \int_{-R}^R \frac{e^{i\omega x}}{1+x^2} dx + \int_0^\pi \frac{e^{i\omega R e^{it}} i R e^{it} dt}{1+R^2 e^{i2t}}$$

But:

$$I_1 = \lim_{R \rightarrow \infty} \int_0^\pi \frac{e^{i\omega R e^{it}} i R e^{it} dt}{1+R^2 e^{i2t}} \rightarrow 0$$

Proof:

$$e^{i\omega R e^{it}} = e^{i\omega R(\cos t + i \sin t)} = e^{i\omega R \cos t} e^{-\omega R \sin t}$$

$$\lim_{R \rightarrow \infty} |e^{i\omega R e^{it}}| = \lim_{R \rightarrow \infty} |e^{i\omega R \cos t}| \times |e^{-\omega R \sin t}| = 1 \times 0 = 0, \quad t \in [0, \pi]$$

On the other hand, in the integral I_1 the remaining factor of the integrand tends to zero as R tends to infinity. So the limit of I_1 as $R \rightarrow \infty$ is zero.

Therefore

$$\pi e^{-\omega} = \oint_C \frac{e^{i\omega z}}{1+z^2} dz = \int_{-R}^R \frac{e^{i\omega x}}{1+x^2} dx = \int_{-R}^R \frac{(\cos x + i \sin x) dx}{1+x^2} = \int_{-R}^R \frac{\cos x dx}{1+x^2} + i \int_{-R}^R \frac{\sin x dx}{1+x^2}$$

The second integral is zero. Why?

$$\int_{-\infty}^{\infty} \frac{\cos \omega x}{1+x^2} dx = \pi e^{-\omega}$$

Example 5.23

Evaluate the integral $I = \int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)(x^2 + 4)}$ using Cauchy's integral formula.

$$\frac{1}{(x^2 + 1)(x^2 + 4)} = \frac{A}{x^2 + 1} + \frac{B}{x^2 + 4} = \frac{1/3}{x^2 + 1} + \frac{-1/3}{x^2 + 4}$$

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2 + 1)(x^2 + 4)} = \frac{1}{3} \int_{-\infty}^{\infty} \frac{dx}{x^2 + 1} - \frac{1}{3} \int_{-\infty}^{\infty} \frac{dx}{x^2 + 4}$$

First integral: $I_1 = \int_{-\infty}^{\infty} \frac{dx}{x^2 + 1}$ and $g_1(z) = \frac{1}{z^2 + 1}$:

$$\oint_C g_1(z) dz = \oint_C \frac{1}{z^2 + 1} dz = \oint_C \left[\frac{1}{z + i} \right] dz = 2\pi i \frac{1}{i + i} = \pi$$

$$\oint_C \frac{1}{z^2 + 1} dz = \pi = \int_{-R}^R \frac{dx}{1 + x^2} + \int_0^\pi \frac{iRe^{it} dt}{1 + R^2 e^{i2t}}$$

$$\lim_{R \rightarrow \infty} \oint_C \frac{1}{z^2 + 1} dz = \int_{-R}^R \frac{dx}{1 + x^2} + 0 = \pi \implies I_1 = \pi$$

Second integral: $I_2 = \int_{-\infty}^{\infty} \frac{dx}{x^2 + 4}$ and $g_2(z) = \frac{1}{z^2 + 4}$:

$$\oint_C g_2(z) dz = \oint_C \frac{1}{z^2 + 4} dz = \oint_C \left[\frac{1}{z + 2i} \right] dz = 2\pi i \frac{1}{2i + 2i} = \frac{\pi}{2}$$

$$\oint_C \frac{1}{z^2 + 4} dz = \frac{\pi}{2} = \int_{-R}^R \frac{dx}{x^2 + 4} + \int_0^\pi \frac{iRe^{it} dt}{R^2 e^{i2t} + 4}$$

$$\lim_{R \rightarrow \infty} \oint_C \frac{1}{z^2 + 4} dz = \int_{-R}^R \frac{dx}{x^2 + 4} + 0 = \frac{\pi}{2} \implies I_2 = \frac{\pi}{2}$$

As a result

$$I = \frac{1}{3}I_1 - \frac{1}{3}I_2 = \frac{1}{3}\pi - \frac{1}{3} \times \frac{\pi}{2} = \frac{\pi}{6}$$

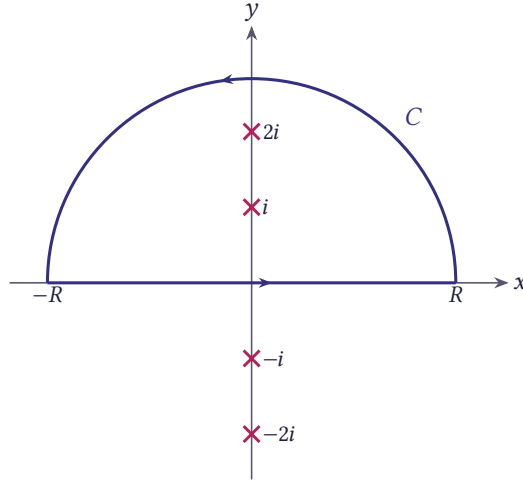


Figure 5.16 The semicircular path; the poles are at $z = \pm i$ and $z = \pm 2i$

Example 5.24

Evaluate the integral $I = \int_0^{\infty} \frac{dx}{x^4 + 1}$ using Cauchy's integral formula.

$$g(z) = \frac{1}{z^4 + 1}, \quad z^4 + 1 = 0 \implies z_1 = e^{i\pi/4}, \quad z_2 = e^{i3\pi/4}, \quad z_3 = e^{i5\pi/4}, \quad z_4 = e^{i7\pi/4}$$

Only the two poles z_1 and z_2 lie inside the path C :

$$\begin{aligned} \oint_C g(z) dz &= \oint_C \frac{1}{z^4 + 1} dz = \oint_{C_1} \frac{1}{z^4 + 1} dz + \oint_{C_2} \frac{1}{z^4 + 1} dz \\ &= \oint_{C_1} \frac{1}{(z - z_2)(z - z_3)(z - z_4)} dz + \oint_{C_2} \frac{1}{(z - z_1)(z - z_3)(z - z_4)} dz \\ &= 2\pi i \frac{1}{(z - z_2)(z - z_3)(z - z_4)} \Big|_{z=z_1} + 2\pi i \frac{1}{(z - z_1)(z - z_3)(z - z_4)} \Big|_{z=z_2} \\ &= 2\pi i \frac{1}{\underbrace{(e^{i\pi/4} - e^{i3\pi/4})}_{\sqrt{2}} \underbrace{(e^{i\pi/4} - e^{i5\pi/4})}_{\sqrt{2}+i\sqrt{2}} \underbrace{(e^{i\pi/4} - e^{i7\pi/4})}_{i\sqrt{2}}} \\ &\quad + 2\pi i \frac{1}{\underbrace{(e^{i3\pi/4} - e^{i\pi/4})}_{-\sqrt{2}} \underbrace{(e^{i3\pi/4} - e^{i5\pi/4})}_{i\sqrt{2}} \underbrace{(e^{i3\pi/4} - e^{i7\pi/4})}_{-\sqrt{2}+i\sqrt{2}}} = \frac{\pi}{\sqrt{2}} \end{aligned}$$

On the other hand

$$\begin{aligned} \oint_C \frac{1}{z^4 + 1} dz &= \frac{\pi}{\sqrt{2}} = \int_{-R}^R \frac{dx}{x^4 + 1} + \int_0^{\pi} \frac{iRe^{it} dt}{R^4 e^{i4t} + 1} \\ \lim_{R \rightarrow \infty} \oint_C \frac{1}{z^4 + 1} dz &= \int_{-\infty}^{\infty} \frac{dx}{x^4 + 1} + \underbrace{\lim_{R \rightarrow \infty} \int_0^{\pi} \frac{iRe^{it} dt}{R^4 e^{i4t} + 1}}_0 = \frac{\pi}{\sqrt{2}} \\ \int_{-\infty}^{\infty} \frac{dx}{x^4 + 1} &= \frac{\pi}{\sqrt{2}} \implies \int_0^{\infty} \frac{dx}{x^4 + 1} = \frac{\pi}{2\sqrt{2}} \end{aligned}$$

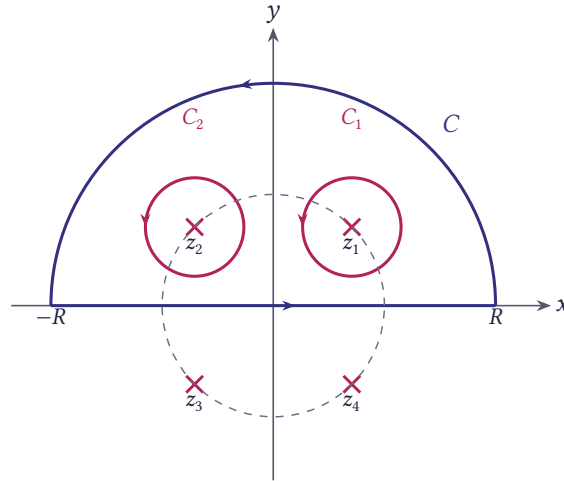


Figure 5.17 The semicircular path and the circles C_1 and C_2 around the poles z_1 and z_2 ; the poles z_3 and z_4 lie in the lower half-plane

5.8 Cauchy's Theorem with Several Non-Analytic Points

Theorem 5.25

If the function $f(z)$ is analytic at all points of the domain D except the points $z_1, z_2, z_3, \dots, z_n$ inside C , then

$$\oint_C f(z) dz = \oint_{C_1} f(z) dz + \oint_{C_2} f(z) dz + \dots + \oint_{C_n} f(z) dz$$

where C_k is a circle centred at z_k that lies inside C and does not contain any of the other points $z_1, \dots, z_n$.

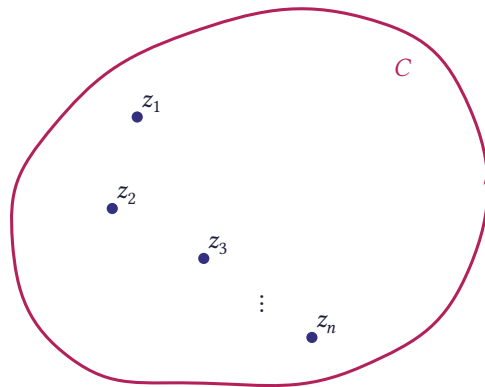


Figure 5.18 A path C and the non-analytic points $z_1, \dots, z_n$ inside it

Proof.

Build the path Γ from C (counterclockwise), the small circles $C_1, \dots, C_n$ (clockwise) and the segments connecting them to C ; the integral along each connecting segment is traversed twice in opposite directions and cancels. The function is analytic inside Γ , so

$$\oint_{\Gamma} f(z) dz = 0 \implies \oint_C f(z) dz + \oint_{C_1^c} f(z) dz + \dots + \oint_{C_n^c} f(z) dz = 0$$

$$\oint_C f(z) dz = -\oint_{C_1^\cup} f(z) dz - \dots - \oint_{C_n^\cup} f(z) dz = \oint_{C_1} f(z) dz + \dots + \oint_{C_n} f(z) dz$$

where $C_k^\cup$ is the circle C_k traversed clockwise and C_k the same circle traversed counterclockwise.

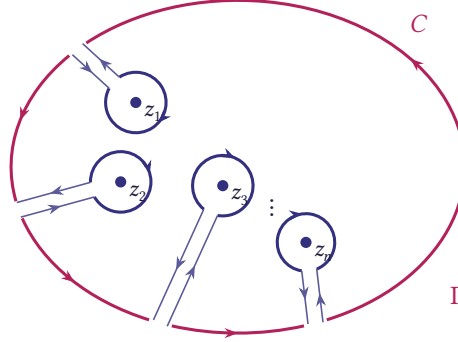


Figure 5.19 The path Γ for the proof: C , the circles $C_1, \dots, C_n$ and the connecting segments

5.9 Derivatives of an Analytic Function

Theorem 5.26 ■ Cauchy's integral formula for the derivative

If the function $f(z)$ is analytic in the simply connected domain D and z_0 is a point inside C , then

$$\frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz = f'(z_0)$$

Proof.

$$f'(z_0) = \lim_{\Delta t \rightarrow 0} \frac{f(z_0 + \Delta t) - f(z_0)}{\Delta t}, \quad f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz, \quad f(z_0 + \Delta t)$$

$$= \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0 - \Delta t} dz$$

$$\begin{aligned} \frac{f(z_0 + \Delta t) - f(z_0)}{\Delta t} &= \frac{1}{2\pi i \Delta t} \left\{ \oint_C \frac{f(z)}{z - z_0 - \Delta t} dz - \oint_C \frac{f(z)}{z - z_0} dz \right\} \\ &= \frac{1}{2\pi i \Delta t} \oint_C f(z) \left[\frac{1}{z - z_0 - \Delta t} - \frac{1}{z - z_0} \right] dz \\ &= \frac{1}{2\pi i \Delta t} \oint_C f(z) \left[\frac{\Delta t}{(z - z_0 - \Delta t)(z - z_0)} \right] dz \\ &= \frac{1}{2\pi i} \oint_C f(z) \left[\frac{1}{(z - z_0 - \Delta t)(z - z_0)} \right] dz \end{aligned}$$

$$\begin{aligned} f'(z_0) &= \lim_{\Delta t \rightarrow 0} \frac{f(z_0 + \Delta t) - f(z_0)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{1}{2\pi i} \oint_C f(z) \left[\frac{1}{(z - z_0 - \Delta t)(z - z_0)} \right] dz \\ &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz \end{aligned}$$

Theorem 5.27 ■ Generalization of Cauchy's integral formula

If the function $f(z)$ is analytic in the simply connected domain D and z_0 is a point inside C , then

$$\frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz = f^{(n)}(z_0)$$

Exercise 5.28

Prove it by mathematical induction.

Example 5.29

Evaluate the following integral. (C is the unit circle.)

$$I = \oint_C \frac{z^2}{(2z - 1)^2} dz$$

$$I = \oint_C \frac{z^2}{4(z - 1/2)^2} dz = \frac{1}{4} \oint_C \frac{z^2}{(z - 1/2)^2} dz = \frac{1}{4} \times 2\pi i \times \left. \frac{d}{dz}(z^2) \right|_{z=1/2} = \frac{1}{4} \times 2\pi i \times (2z) \Big|_{z=1/2}$$

$$= \frac{\pi}{2} i$$

($z = 1/2$ lies inside the unit circle.)

Example 5.30

Evaluate the following integral. (C is the unit circle.)

$$I = \oint_C \frac{e^{-z} \sin z}{z^2} dz$$

$$f(z) = e^{-z} \sin z, \quad z_0 = 0, \quad f'(z) = -e^{-z} \sin z + e^{-z} \cos z, \quad f'(z_0) = f'(0) = 1$$

$$I = 2\pi i f'(z_0) = 2\pi i$$

Exercise 5.31 ■ Problem

If $f(z)$ is analytic in the whole z -plane and bounded in the whole plane, i.e. $|f(z)| < M$, prove that $f(z) = k$, $k \in \mathbb{C}$.

Choose the path C to be a circle of radius R centred at z_0 , and let R tend to infinity.

$$\begin{aligned}
 f'(z_0) &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz \\
 |f'(z_0)| &= \left| \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^2} dz \right| \leq \frac{1}{2\pi} \oint_C \frac{|f(z)|}{|z - z_0|^2} |dz| \leq \frac{1}{2\pi} \frac{M}{R^2} \oint_C |dz| = \frac{1}{2\pi} \frac{M}{R^2} \cdot 2\pi R \\
 &= \frac{M}{R} \rightarrow 0 \quad (R \rightarrow \infty) \\
 \implies |f'(z_0)| = 0 &\implies f'(z) = 0 \implies f(z) = k
 \end{aligned}$$

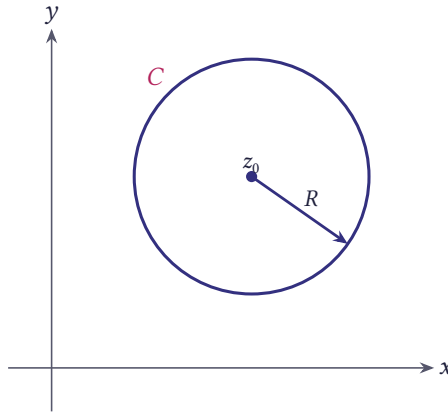


Figure 5.20 A circle C with centre z_0 and radius R

Exercise 5.32

Evaluate the following integrals:

1. $\oint_C \frac{z^2}{(2z - 1)^4} dz$, C : the unit circle
2. $\int_{1+i}^{1+3i} e^z \sin z \, dz$
3. $\int_{1+i}^{2+2i} \operatorname{Re}\{z^2\} \, dz$ along a straight-line path
4. $\oint_C \frac{e^{\cos z} \sin z}{1 + z^6} dz$ along the path below

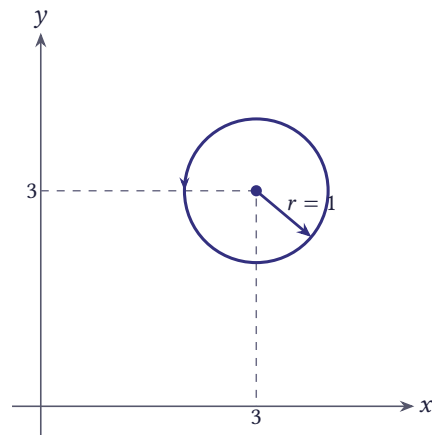


Figure 5.21 The path of exercise 4: a circle with centre $3 + 3i$ and radius $r = 1$